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THINK MERIT | THINK TRANSPARENCY | THINK SASTRA

OPERATING SYSTEMS LABORATORY

Course Code: CSE309R01

Semester: V

Lab Manual

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**SHANMUGHA ARTS, SCIENCE, TECHNOLOGY AND RESEARCH
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Exercise No.9. Simulating Reader-Writer problem

Write a program simulate reader-writer problem

Objectives

To write a code to solve the readers writer's problem based on reader priority and writer priority solution

Prerequisite / Tools Required (optional)

Knowledge of Concurrency, Mutual exclusion, Synchronization and Reader writer problem

Procedure / Algorithm

- ☐ Create a reader process
- ☐ Create a writer process
- ☐ Implement necessary semaphores
- ☐ Implement the programs giving reader priority and writer priority

CODE:

READER WRITER MAIN:

```
#include<stdio.h>
#include<stdlib.h>
#include<sys/ipc.h>
#include<sys/shm.h>
#include<string.h>
#include<unistd.h>
#define SHMSZ 1024
int main() {
    key_t key1, key2;
    int shmid;
    int sec_id;
    char *shm, *k, *s; // Data
    char *ssm, *j, *t;
    key2 = 3400; // Pattern
    key1 = 3415; // Data
    if ((shmid = shmget(key1, SHMSZ, IPC_CREAT | 0666)) < 0) {
        perror("shmget");
        exit(1);
    }
```

```

if ((shm = shmat(shmid, NULL, 0)) == (char *) -1) {
perror("shmat");
exit(1);
}
//*****
if ((sec_id = shmget(key2, SHMSZ, IPC_CREAT | 0666)) < 0) {
perror("shmget");
exit(1);
}
if ((ssm = shmat(sec_id, NULL, 0)) == (char *) -1) {
perror("shmat");
exit(1);
}
t = ssm;
s = shm;
char pattern[10];
printf("\nEnter the pattern in caps: ");
fgets(pattern, 10, stdin); // Will accept till ENTER key is hit...
int len = strlen(pattern); // Corrected length calculation
for (int i = 0; i < len; i++) {
*t = pattern[i];
*s = pattern[i];
t++;
s++;
}
// Print the pattern written to shared memory
printf("\nPattern written to shm mem of main : %s\nPattern written to wwp & rwp shm mem: ", ssm);
for(char* i=shm; *i != '\0'; i++)
printf("%c", *i);
putchar('\n');
int choice;
printf("1. READER PRIORITY\n2. WRITER PRIORITY\n3. BASED ON FIRST\n4. AS PER GIVEN\n");
printf("Enter the priority (1/2/3/4): ");
scanf("%d", &choice);
if (choice == 1)
{
// Reader Priority
char first = *ssm;
printf("\nThe first character is: %c", first);
int rcount = 0;
int wcount = 0;
for (k = ssm; *k != '\0'; k++) {
if (*k == 'R')
rcount++;
else if (*k == 'W')

```

```

wcount++;
}
printf("\nThe no. of readers is: %d", rcount);
printf("\nThe no. of writers is: %d\n", wcount);
for (int i = 0; i < rcount; i++) {
system("gnome-terminal -- ./r.out");
sleep(5);
}
for (int i = 0; i < wcount; i++) {
system("gnome-terminal -- ./w.out");
sleep(10);
}
}
else if (choice == 2)
{
// Writer Priority
char first = *ssm;
printf("\nThe first character is: %c", first);
int rcount = 0;
int wcount = 0;
for (k = ssm; *k != '\0'; k++) {
if (*k == 'R')
rcount++;
else if (*k == 'W')
wcount++;
}
printf("\nThe no. of readers is: %d", rcount);
printf("\nThe no. of writers is: %d\n", wcount);
// Open writer and reader processes in separate terminals
for (int i = 0; i < wcount; i++) {
system("gnome-terminal -- ./w.out");
sleep(10);
}
for (int i = 0; i < rcount; i++) {
system("gnome-terminal -- ./r.out");
sleep(5);
}
}
else if (choice == 3)
{
// based on first
char first = *ssm;
printf("\nThe first character is: %c", first);
char *h;
h = ssm;
if (*h == 'R') {
int rcount = 0;

```

```

int wcount = 0;
for (k = h; *k != '\0'; k++) {
    if (*k == 'R')
        rcount++;
    else if (*k == 'W')
        wcount++;
} printf("\nThe no. of readers is: %d", rcount);
printf("\nThe no. of writers is: %d\n", wcount);
sleep(10);
for (int i = 0; i < rcount; i++) {
    system("gnome-terminal -- ./r.out");
    sleep(5);
}
for (int i = 0; i < wcount; i++) {
    system("gnome-terminal -- ./w.out");
    sleep(10);
}
} else if (*h == 'W') {
    int rcount = 0;
    int wcount = 0;
    for (k = h; *k != '\0'; k++) {
        if (*k == 'R')
            rcount++;
        else if (*k == 'W')
            wcount++;
    } printf("\nThe no. of readers is: %d", rcount);
    printf("\nThe no. of writers is: %d\n", wcount);
    sleep(5);
    for (int i = 0; i < wcount; i++) {
        system("gnome-terminal -- ./w.out");
        sleep(10);
    }
    for (int i = 0; i < rcount; i++) {
        system("gnome-terminal -- ./r.out");
        sleep(5);
    }
}
}
} else if ( choice == 4 )
{
    // as per given
    char f=*ssm;
    char *h=ssm;
    int rcount = 0;
    int wcount = 0;
    for (k = h; *k != '\0'; k++) {
        if (*k == 'R')

```

```

rcount++;
else if (*k == 'W')
wcount++;
}
printf("\nThe first character is : %c",f);
printf("\nThe no. of readers is: %d", rcount);
printf("\nThe no. of writers is: %d\n", wcount);
while(*h != '\0')
{
if(*h == 'R')
{
system("gnome-terminal -- ./r.out");
sleep(5);
}
else if(*h == 'W')
{
system("gnome-terminal -- ./w.out");
sleep(10);
}
h++;
}
}
else {
printf("Invalid choice.\n");
}
// Detach and remove shared memory segments
shmdt(shm);
shmdt(ssm);
shmctl(shmid, IPC_RMID, NULL);
shmctl(sec_id, IPC_RMID, NULL);
return 0;
}

```

READER:

```

#include<stdio.h>
#include<stdlib.h>
#include<sys/types.h>
#include<sys/wait.h>
#include<unistd.h>
#include <sys/shm.h>
#define SHMSZ 1024
int main() {
int shmid;
key_t key;
key = 3415;
char *shm, *s, *k;
if ((shmid = shmget(key, SHMSZ, 0666)) < 0) {

```

```

perror("shmget");
exit(1);
}
if ((shm = shmat(shmid, NULL, 0)) == (char *) -1) {
perror("shmat");
exit(1);
}
char flag = *shm;
s = shm;
printf("\nThe data at the first location of the Shared Memory is:%c\n", flag);
printf("The values in the shared memory : %s\n",shm);
// for (k = shm; *k != '\0'; k++) {
// printf("%c", *k);
// }
putchar('\n');
sleep(5);
printf("\nThe Read process of the Reader is complete!!!\n");
return 0;
}

```

WRITER:

```

#include<stdio.h>
#include<stdlib.h>
#include<sys/types.h>
#include<sys/wait.h>
#include<unistd.h>
#include <sys/shm.h>
#include<stdbool.h>
#define SHMSZ 1024
#define MAX_LIMIT 1024
int main() {
int shmid;
key_t key;
key = 3415;
char *shm, *s, *k;
if ((shmid = shmget(key, SHMSZ, 0666)) < 0) {
perror("shmget");
exit(1);
}
if ((shm = shmat(shmid, NULL, 0)) == (char *) -1) {
perror("shmat");
exit(1);
}
s = shm;
printf("\n %d is the Write process ID that is writing into Shared Memory...\n", getpid());
while (*s != '\0')
s++;

```

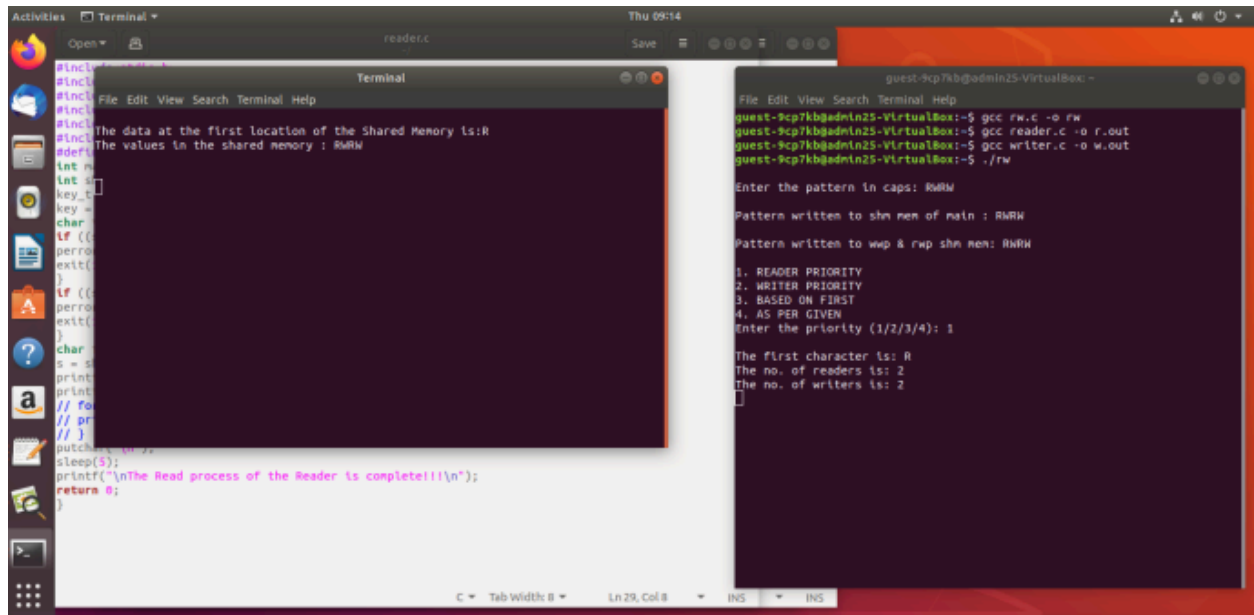
```

char x = *s; // The data here should be null
printf("\n The Data here should be null ( coz this is the end of shm) : %c \n", x);
char str[MAX_LIMIT];
int choice;
do {
printf("\n Enter the String to be entered into the Shared Memory: ");
fgets(str, MAX_LIMIT, stdin); // Will accept input till ENTER key is hit...
printf("\nAre you sure the string you want to enter is: %s",str);
printf("\nIf YES -> 1 | NO -> 0: ");
scanf("%d", &choice);
getchar(); // Consume the newline character left by scanf
} while (choice != 1);
printf("The String that will be entered into the SHM is: %s\n",str);
// Copy the string into shared memory
for (int i = 0; i < MAX_LIMIT; i++) {
*s++ = str[i]; // After the last memory location, s will point to NULL
if (str[i] == '\0')
break;
}
sleep(5); // Simulate some processing time
printf("\nThe Write process for %d Process is complete!!!\n",getpid());
printf("Contents of the shared memory after writing: ");
for (k = shm; *k != '\0'; k++) {
printf("%c", *k);
}
putchar('\n');
return 0;
}

```

Output

READER PRIORITY:



```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <string.h>
#include <sys/types.h>
#include <sys/shm.h>
#include <sys/ipc.h>
#include <fcntl.h>
#include <pthread.h>

#define SHM_SIZE 1024
#define SHM_NAME "/shm"
#define WWP_SIZE 1024
#define WWP_NAME "/wwp"

int main()
{
    int shmid;
    int wwpid;
    key_t key_t;
    char *shm;
    char *wwp;
    if ((shmid = shm_open(SHM_NAME, O_RDWR, 0666)) < 0)
        perror("shm_open");
    if ((wwpid = shm_open(WWP_NAME, O_RDWR, 0666)) < 0)
        perror("shm_open");
    shm = mmap(0, SHM_SIZE, PROT_READ|PROT_WRITE, MAP_SHARED, shmid, 0);
    wwp = mmap(0, WWP_SIZE, PROT_READ|PROT_WRITE, MAP_SHARED, wwpid, 0);
    printf("The data at the first location of the Shared Memory is:R\n");
    printf("The values in the shared memory : RWRW\n");
    return 0;
}
```

```
guest-9cp7kb@admin25-VirtualBox:~$ gcc rw.c -o rw
guest-9cp7kb@admin25-VirtualBox:~$ gcc reader.c -o r.out
guest-9cp7kb@admin25-VirtualBox:~$ gcc writer.c -o w.out
guest-9cp7kb@admin25-VirtualBox:~$ ./rw

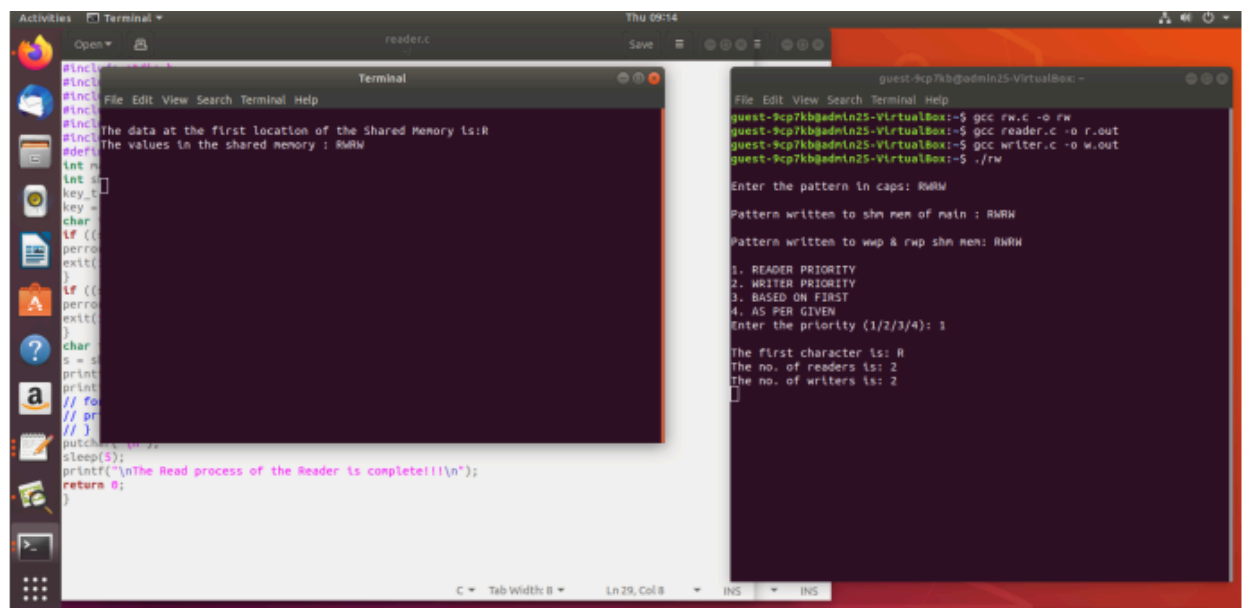
Enter the pattern in caps: RWRW

Pattern written to shm mem of main : RWRW

Pattern written to wwp & rwp shm mem: RWRW

1. READER PRIORITY
2. WRITER PRIORITY
3. BASED ON FIRST
4. AS PER GIVEN
Enter the priority (1/2/3/4): 1

The first character is: R
The no. of readers is: 2
The no. of writers is: 2
```



```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <string.h>
#include <sys/types.h>
#include <sys/shm.h>
#include <sys/ipc.h>
#include <fcntl.h>
#include <pthread.h>

#define SHM_SIZE 1024
#define SHM_NAME "/shm"
#define WWP_SIZE 1024
#define WWP_NAME "/wwp"

int main()
{
    int shmid;
    int wwpid;
    key_t key_t;
    char *shm;
    char *wwp;
    if ((shmid = shm_open(SHM_NAME, O_RDWR, 0666)) < 0)
        perror("shm_open");
    if ((wwpid = shm_open(WWP_NAME, O_RDWR, 0666)) < 0)
        perror("shm_open");
    shm = mmap(0, SHM_SIZE, PROT_READ|PROT_WRITE, MAP_SHARED, shmid, 0);
    wwp = mmap(0, WWP_SIZE, PROT_READ|PROT_WRITE, MAP_SHARED, wwpid, 0);
    printf("The data at the first location of the Shared Memory is:R\n");
    printf("The values in the shared memory : RWRW\n");
    return 0;
}
```

```
guest-9cp7kb@admin25-VirtualBox:~$ gcc rw.c -o rw
guest-9cp7kb@admin25-VirtualBox:~$ gcc reader.c -o r.out
guest-9cp7kb@admin25-VirtualBox:~$ gcc writer.c -o w.out
guest-9cp7kb@admin25-VirtualBox:~$ ./rw

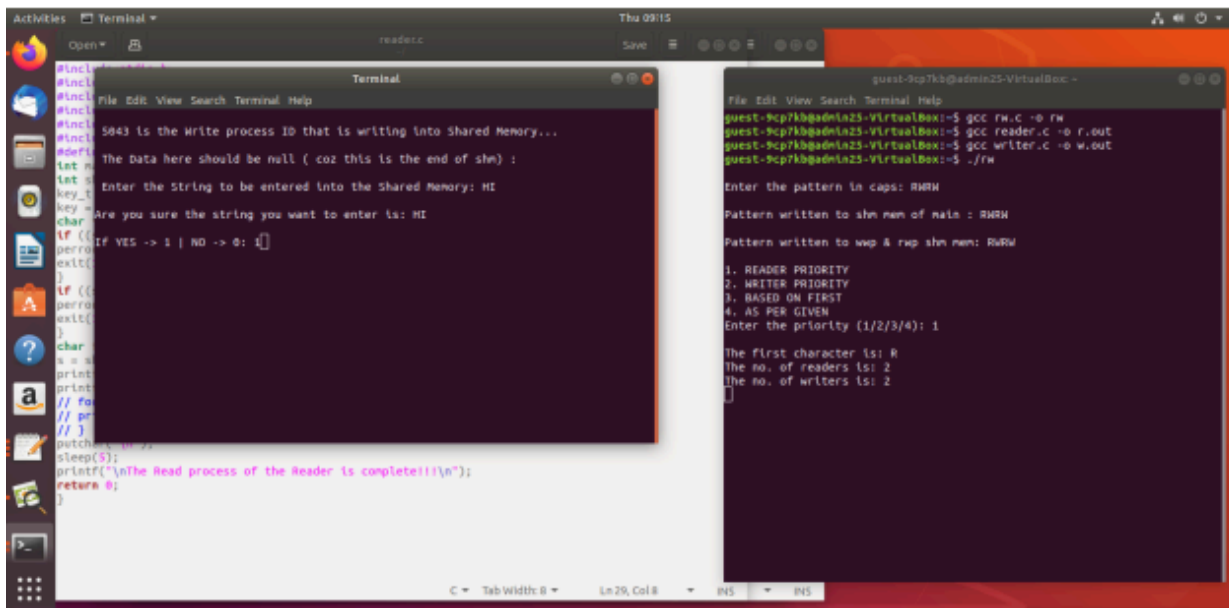
Enter the pattern in caps: RWRW

Pattern written to shm mem of main : RWRW

Pattern written to wwp & rwp shm mem: RWRW

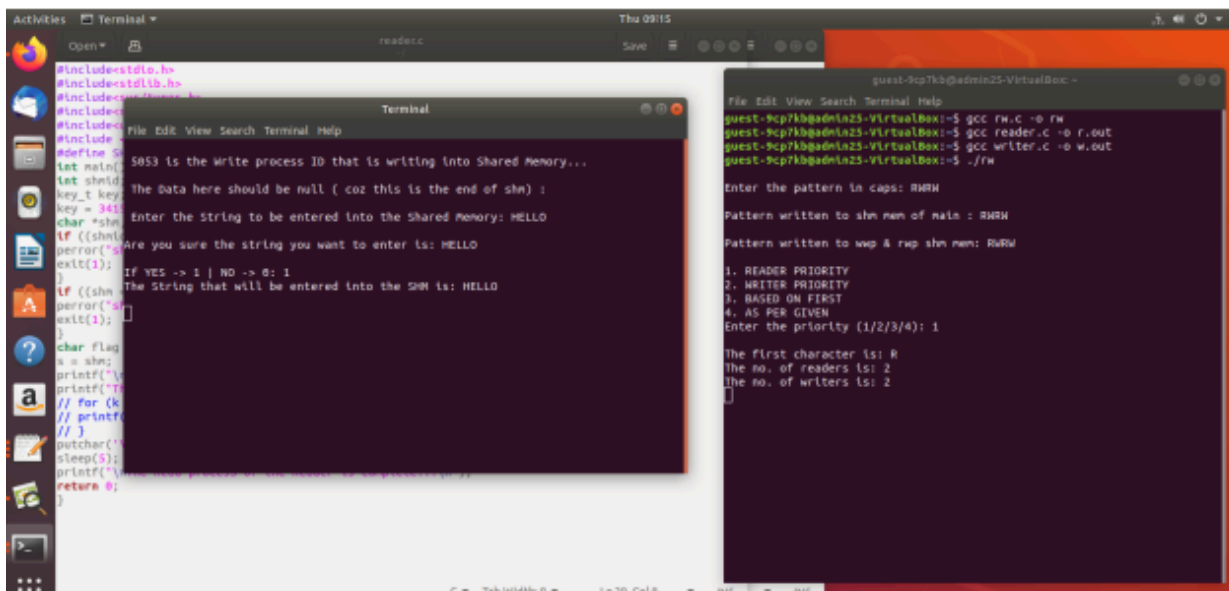
1. READER PRIORITY
2. WRITER PRIORITY
3. BASED ON FIRST
4. AS PER GIVEN
Enter the priority (1/2/3/4): 1

The first character is: R
The no. of readers is: 2
The no. of writers is: 2
```



```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/shm.h>
#include <sys/ipc.h>
#define SHM_SIZE 1024
int main()
{
    int shmid;
    char *shm;
    key_t key = 365;
    char *shm;
    if ((shmid = shmget(key, SHM_SIZE, IPC_CREAT | 0666)) < 0)
    {
        perror("shmget");
        exit(1);
    }
    if ((shm = shmat(shmid, NULL, 0)) == NULL)
    {
        perror("shmat");
        exit(1);
    }
    printf("5043 is the Write process ID that is writing into Shared Memory...\n");
    printf("The Data here should be null ( coz this is the end of shm) :\n");
    printf("Enter the String to be entered into the Shared Memory: HI\n");
    printf("Are you sure the string you want to enter is: HI\n");
    if (YES -> 1 | NO -> 0: 1)
    {
        if ((shm = shmat(shmid, NULL, 0)) == NULL)
        {
            perror("shmat");
            exit(1);
        }
        char *shm;
        a = shm;
        printf("The Read process of the Reader is complete!!\n");
        return 0;
    }
}
```

```
guest-9cp7kb@admin25-VirtualBox:~$ gcc rw.c -o r.w
guest-9cp7kb@admin25-VirtualBox:~$ gcc reader.c -o r.out
guest-9cp7kb@admin25-VirtualBox:~$ gcc writer.c -o w.out
guest-9cp7kb@admin25-VirtualBox:~$ ./r.w
Enter the pattern in caps: RWRW
Pattern written to shm mem of main : RWRW
Pattern written to wwp & rwp shm mem: RWRW
1. READER PRIORITY
2. WRITER PRIORITY
3. BASED ON FIRST
4. AS PER GIVEN
Enter the priority (1/2/3/4): 1
The first character is: R
The no. of readers is: 2
The no. of writers is: 2
```



```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/shm.h>
#include <sys/ipc.h>
#define SHM_SIZE 1024
int main()
{
    int shmid;
    char *shm;
    key_t key = 365;
    char *shm;
    if ((shmid = shmget(key, SHM_SIZE, IPC_CREAT | 0666)) < 0)
    {
        perror("shmget");
        exit(1);
    }
    if ((shm = shmat(shmid, NULL, 0)) == NULL)
    {
        perror("shmat");
        exit(1);
    }
    printf("5053 is the Write process ID that is writing into Shared Memory...\n");
    printf("The Data here should be null ( coz this is the end of shm) :\n");
    printf("Enter the String to be entered into the Shared Memory: HELLO\n");
    printf("Are you sure the string you want to enter is: HELLO\n");
    if (YES -> 1 | NO -> 0: 1)
    {
        if ((shm = shmat(shmid, NULL, 0)) == NULL)
        {
            perror("shmat");
            exit(1);
        }
        char *shm;
        a = shm;
        printf("The Read process of the Reader is complete!!\n");
        return 0;
    }
}
```

```
guest-9cp7kb@admin25-VirtualBox:~$ gcc rw.c -o r.w
guest-9cp7kb@admin25-VirtualBox:~$ gcc reader.c -o r.out
guest-9cp7kb@admin25-VirtualBox:~$ gcc writer.c -o w.out
guest-9cp7kb@admin25-VirtualBox:~$ ./r.w
Enter the pattern in caps: RWRW
Pattern written to shm mem of main : RWRW
Pattern written to wwp & rwp shm mem: RWRW
1. READER PRIORITY
2. WRITER PRIORITY
3. BASED ON FIRST
4. AS PER GIVEN
Enter the priority (1/2/3/4): 1
The first character is: R
The no. of readers is: 2
The no. of writers is: 2
```

WRITER PRIORITY:

```
Activities Terminal Thu 09:28
Open / Open Save
Terminal
File Edit View Search Terminal Help
5762 is the Write process ID that is writing into Shared Memory...
The Data here should be null ( coz this is the end of shm ) :
Enter the String to be entered into the Shared Memory: HI
Are you sure the string you want to enter is: HI
If YES -> 1 | NO -> 0: 1
The String that will be entered into the SHM is: HI
Contents of the shared memory after writing:
HI
The first character is: H
The no. of readers is: 2
The no. of writers is: 2

guest-9cp7kb@admin25-VirtualBox:~$ gcc rw.c -o rw
guest-9cp7kb@admin25-VirtualBox:~$ gcc reader.c -o r.out
guest-9cp7kb@admin25-VirtualBox:~$ gcc writer.c -o w.out
guest-9cp7kb@admin25-VirtualBox:~$ ./rw
Enter the pattern in caps: RWRW
Pattern written to shn mem of main : RWRW
Pattern written to wwp & rwp shn mem: RWRW
1. READER PRIORITY
2. WRITER PRIORITY
3. BASED ON FIRST
4. AS PER GIVEN
Enter the priority (1/2/3/4): 2
The first character is: R
The no. of readers is: 2
The no. of writers is: 2
```

```
Activities Terminal Thu 09:28
Open / Open Save
Terminal
File Edit View Search Terminal Help
5715 is the Write process ID that is writing into Shared Memory...
The Data here should be null ( coz this is the end of shm ) :
Enter the String to be entered into the Shared Memory: HELLO
Are you sure the string you want to enter is: HELLO
If YES -> 1 | NO -> 0: 1
The String that will be entered into the SHM is: HELLO
Contents of the shared memory after writing:
HELLO
The first character is: H
The no. of readers is: 2
The no. of writers is: 2

guest-9cp7kb@admin25-VirtualBox:~$ gcc rw.c -o rw
guest-9cp7kb@admin25-VirtualBox:~$ gcc reader.c -o r.out
guest-9cp7kb@admin25-VirtualBox:~$ gcc writer.c -o w.out
guest-9cp7kb@admin25-VirtualBox:~$ ./rw
Enter the pattern in caps: RWRW
Pattern written to shn mem of main : RWRW
Pattern written to wwp & rwp shn mem: RWRW
1. READER PRIORITY
2. WRITER PRIORITY
3. BASED ON FIRST
4. AS PER GIVEN
Enter the priority (1/2/3/4): 2
The first character is: R
The no. of readers is: 2
The no. of writers is: 2
```

```
#include <stdio.h>
#include <unistd.h>
#include <sys/mman.h>
#include <sys/types.h>
#include <fcntl.h>
#include <pthread.h>
#include <stdlib.h>
#include <string.h>
#include <time.h>

#define SHM_SIZE 1024
#define SHM_NAME "/shm"
#define READERS 2
#define WRITERS 2

int main()
{
    int fd;
    char *shm;
    pthread_t readers[READERS];
    pthread_t writers[WRITERS];

    fd = shm_open(SHM_NAME, O_CREAT | O_RDWR, 0666);
    if (fd == -1)
    {
        perror("shm_open");
        return 1;
    }

    shm = mmap(0, SHM_SIZE, PROT_READ | PROT_WRITE, MAP_SHARED, fd, 0);
    if (shm == MAP_FAILED)
    {
        perror("mmap");
        return 1;
    }

    printf("The data at the first location of the Shared Memory is:R\n");
    printf("The values in the shared memory : RWRW\n");

    for (int i = 0; i < READERS; i++)
    {
        pthread_create(&readers[i], NULL, reader, (void *)i);
    }

    for (int i = 0; i < WRITERS; i++)
    {
        pthread_create(&writers[i], NULL, writer, (void *)i);
    }

    for (int i = 0; i < READERS; i++)
    {
        pthread_join(readers[i], NULL);
    }

    for (int i = 0; i < WRITERS; i++)
    {
        pthread_join(writers[i], NULL);
    }

    printf("The data at the first location of the Shared Memory is:R\n");
    printf("The values in the shared memory : RWRW\n");

    return 0;
}

// Reader function
void *reader(void *arg)
{
    int id = (int)arg;
    char *shm = (char *)0;
    int fd = shm_open(SHM_NAME, O_CREAT | O_RDWR, 0666);
    shm = mmap(0, SHM_SIZE, PROT_READ | PROT_WRITE, MAP_SHARED, fd, 0);

    sleep(5); // Simulate some processing time
    printf("\n\nThe Write process for %d Process is complete!!!\n", getpid());
    printf("Contents of the shared memory after writing: ");
    for (k = shm; *k != '\0'; k++) {
        printf("%c", *k);
    }
    putchar('\n');
    return 0;
}

// Writer function
void *writer(void *arg)
{
    int id = (int)arg;
    char *shm = (char *)0;
    int fd = shm_open(SHM_NAME, O_CREAT | O_RDWR, 0666);
    shm = mmap(0, SHM_SIZE, PROT_READ | PROT_WRITE, MAP_SHARED, fd, 0);

    Enter the pattern in caps: RWRW

    Pattern written to shm mem of main : RWRW

    Pattern written to wup & rwp shm mem: RWRW

    1. READER PRIORITY
    2. WRITER PRIORITY
    3. BASED ON FIRST
    4. AS PER GIVEN
    Enter the priority (1/2/3/4): 2

    The first character is: R
    The no. of readers is: 2
    The no. of writers is: 2
}
```

FCFS:

```
Activities Terminal Thu 09:37
Open Open Save
Terminal
File Edit View Search Terminal Help
6376 is the Write process ID that is writing into Shared Memory...
The Data here should be null ( coz this is the end of shm ) :
Enter the String to be entered into the Shared Memory: HI
Are you sure the string you want to enter is: HI
If YES -> 1 | NO -> 0: 1
The String that will be entered into the SHM is: HI
sleep(5); // Simulate some processing time
printf("\nThe Write process for %d Process is complete!!!\n",getpid());
printf("Contents of the shared memory after writing: ");
for (k = shm; *k != '\0'; k++) {
    printf("%c", *k);
}
putchar('\n');
return 0;
}

n", getpid());
n", k);

guest-9cp7kb@admin25-VirtualBox: ~
File Edit View Search Terminal Help
guest-9cp7kb@admin25-VirtualBox:~$ ./rw
Enter the pattern in caps: WWRW
Pattern written to shm mem of main : WWRW
Pattern written to wwp & rwp shm mem: WWRW
1. READER PRIORITY
2. WRITER PRIORITY
3. BASED ON FIRST
4. AS PER GIVEN
Enter the priority (1/2/3/4): 4
The first character is : W
The no. of readers is: 2
The no. of writers is: 2
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```

```
Activities Terminal Thu 09:38
Open Open Save
Terminal
File Edit View Search Terminal Help
The data at the first location of the Shared Memory is:W
The values in the shared memory : WWRW
Enter the String to be entered into the Shared Memory: HI
Are you sure the string you want to enter is: HI
If YES -> 1 | NO -> 0: 1
The String that will be entered into the SHM is: HI
sleep(5); // Simulate some processing time
printf("\nThe Write process for %d Process is complete!!!\n",getpid());
printf("Contents of the shared memory after writing: ");
for (k = shm; *k != '\0'; k++) {
    printf("%c", *k);
}
putchar('\n');
return 0;
}

n", getpid());
n", k);

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```



```

wait(mutex)
    readcount++
    if (readcount == 1) wait(wrt) // First reader blocks writers
signal(mutex)

// Reading is performed here...

wait(mutex)
    readcount--
    if (readcount == 0) signal(wrt) // Last reader awakens writers
signal(mutex)

// Writer Process
wait(wrt) // Requires exclusive access
// Writing is performed here...
signal(wrt)

```

Use code with caution.

2) Pseudo-code: Writers with Priority

text

```

// Reader Process
wait(rsem) // Queue up if a writer is waiting
wait(mutex1)
    readcount++
    if (readcount == 1) wait(wrt)
    signal(mutex1)
signal(rsem)

// Reading is performed here...

wait(mutex1)
    readcount--
    if (readcount == 0) signal(wrt)
signal(mutex1)

// Writer Process
wait(mutex2)
    writecount++
    if (writecount == 1) wait(rsem) // First writer blocks incoming readers
signal(mutex2)

wait(wrt)
// Writing is performed here...
signal(wrt)

```

```
wait(mutex2)
    writecount--
    if (writecount == 0) signal(rsem) // Last writer unlocks readers

signal(mutex2)
```

RESULT:

Thus a reader writer problem as specified is implemented and executed